

Study Questions: Cognition — Hidden Layers in Hook and Mind

1. Explain how a crocheted foundation chain functions as an "input layer." What information does it carry, and what information does it lack that will only emerge in the hidden layers above?
2. Choose a stitch pattern you know (shell, bobble, basket weave, granny square, or another). Describe how it creates a hidden representation — an emergent texture that cannot be reduced to the individual stitches that compose it.
3. In a neural network, adding more hidden layers increases the network's capacity for complex representations. In crochet, adding more rows of varied stitch patterns increases the fabric's textural complexity. Where does this analogy hold, and where does it break down?
4. A crocheter working a cable stitch creates a raised twist that spans multiple rows. Explain how this multi-row feature is analogous to a feature that emerges across multiple hidden layers in a deep neural network.
5. Describe the crocheter's multi-level mental model using the five levels discussed in the module (stitch, row, section, project, design). For a project you are currently working on or have recently completed, identify what your mind holds at each level.
6. In Active Inference, a deep generative model generates top-down predictions from abstract levels to concrete levels. Give a specific example: how does the crocheter's project-level belief ("I am making a baby blanket") generate predictions about what the current row should look like?
7. When you frog (unravel) several rows of crochet, you lose the emergent textures those rows created. Compare this to what happens when you reinitialize (reset) several hidden layers of a trained neural network. What is lost in each case?

8. Stitch markers serve as external memory in the crocheter's cognitive system. Identify three different types of information that stitch markers carry (e.g., round beginning, shaping points, pattern repeats) and explain how each offloads cognitive work from the internal generative model.
9. A beginner follows a pattern stitch by stitch but cannot tell whether the project is "on track" by looking at it. An expert can tell at a glance. Using the concept of deep versus shallow generative models, explain this difference.
10. Consider a granny square. The center ring is the input. Each round is a hidden layer. The final round is the output. Trace how the visual pattern of the granny square (diagonal lines, open spaces, cluster corners) emerges layer by layer. At which round does the characteristic granny square pattern become recognizable?
11. In a neural network, "skip connections" (as in ResNet architectures) allow information to jump across layers. Can you find an analogy in crochet? Is there a technique where a stitch or structural element connects to something more than one row below?
12. The module describes teaching crochet as "model transfer." When you teach someone a new stitch, which layer of their generative model are you building? How does this differ from teaching them to "read their work"?
13. A crocheter picks up a half-finished project after a long break and cannot remember where they were. Describe this as a failure of the internal generative model. What information has been lost, and how does the crocheter reconstruct their position in the hierarchy?
14. Compare the cognitive demands of crocheting a simple single-crochet washcloth (a "shallow" project) with crocheting a complex lace shawl (a "deep" project). What additional mental layers does the shawl require?
15. In neural networks, the same hidden layer architecture can be trained for different tasks (image recognition, language translation, etc.) by changing the training data. In crochet, the same stitch repertoire can produce vastly different objects by changing the pattern. Discuss this parallel between network flexibility and craft versatility.

16. Describe how a crochet pattern (a written set of instructions) relates to the weights of a trained neural network. In what sense does the pattern "encode" the information needed to reconstruct the fabric, and in what sense is it incomplete?
17. When a crocheter improvises — working without a written pattern — they rely entirely on their internal generative model. What must be true about the depth and precision of this model for successful improvisation? How does this compare to a generative AI model producing novel outputs?
18. Consider the concept of "information loss" in neural network pooling layers, which reduce spatial resolution to capture higher-level features. Is there an analogous process in crochet where detail is sacrificed for overall shape? Think about gauge changes, stitch simplification, or blocking.
19. A crocheter is working a complex colorwork pattern (tapestry crochet) where multiple colors create a picture in the fabric. How many layers of the mental model must be active simultaneously? What does each layer track?
20. Reflect on your own experience: when you crochet, are you more aware of the stitch level (individual stitches) or the project level (the overall shape)? How does your awareness shift as you gain experience with a particular pattern? Relate this to how attention moves through the layers of a deep generative model.